Diseases of clove

1. Sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)
2. Acute Dieback - *Cryphonectria eugeniae*
3. Sumatra disease or die back - *Ralstonia syzygii*

Diseases of cinnamon

1. Bark canker - *Phytophthora cinnamomi*
2. Leaf spot and die-back - *Colletotrichum gloeosporioides*

Diseases of nutmeg

1. Die back and fruit rot - *Phytophthora* spp. and *Diplodia natalensis*
2. Wilt - *Rosellinia pepo*
3. Leaf spot and shot hole - *Colletotrichum gloeosporioides*

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Infected plant

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

- It is a one of the serious disease of clove.
- **Trees of all ages are affected it is destructive mainly in older trees**
- The first symptom **is a slight chlorosis, an almost imperceptible change when considering individual leaves but noticeable in the mass to a practised observer.**
- This is followed by a slight thinning and loss of turgidity of the leaves and then, within a few days and spectacularly rapid death.
- A mass of russet-brown leaves remains on infected trees, so that these stand out clearly against the dark green foliage of healthy ones

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Two kinds of symptom are apparent, a slow decline and a sudden wilt.

Slow decline

- The slow decline phase is characteristic of younger trees.
- Seedlings appear to be immune
- Saplings 8-10 years old are resistant.
- The decline phase is seen in trees up to about 20 years of age and it may take 6 months before trees succumb.

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Sudden wilt.

- Trees more than 30 years old are susceptible and death (sudden wilt) occurs in a few days from the onset of symptoms.
- On mature trees, slight chlorosis occurs on the leaves which results in a sudden and rapid leaf fall followed by wilting of the plants.
- Complete death of the plants occurs within a few days.
- The roots are affected by the pathogen.

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Root infection

- The rain washes ascospores and pycnidiospores into soil to initiate infection.
- The absorbing roots are infected first, later the infection spreads to fine, fibrous roots results eventually in complete destruction of the root system.
- During **the dry period that follows rain, plants suddenly wilt** because roots cannot meet the demand for water to stems and leaves.

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

- A **few days after death**, **a yellow stain in the wood at the collar.**
- This spreads up the trunk and is generally distributed 9 months later but occurs much earlier in the roots.
- The stain is often separated from healthy wood by a sharp **black zone line and a narrow bluish-grey zone.**
- Tyloses and gum occur in the vessels.
- **Some 3-4 months after death of the tree** the perithecia appear, through cracks in the bark, at ground level, and fructification moves upwards.
- Fruiting can continue for several years on trunk and branches.

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

- The pycnidia, which appear before the perithecia, are abundant on cut, split or broken woody surfaces in moist conditions.

- **Where new plantings of clove are made in areas killed off by sudden death, young trees up to nearly 20 years old exhibit a slow decline** and unthrifty appearance with yellowish and sparse foliage accompanied by a slow but progressive root rot.

- Seedlings are immune from infection by *V. eugeniae* and plants up to 4 years old are very resistant

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Pathogen: PS: *Valsa eugeniae* (**IPS:** *Cytospora eugeniae*)

- Cytospora species are ubiquitous pathogens of numerous woody plants, causing dieback and wood cankers in **agronomic crops, timber trees and wildland trees**
- The perithecia and pycnidia are produced in large numbers.
- The ascospores are formed during the rainy season and are washed down to the soil where they infect the roots.
- The rain washes ascospores and pycnidiospores into soil to initiate infection.

Clove sudden death - *Valsa eugeniae* (IPS :*Cytospora eugeniae*)

Pathogen: PS: *Valsa eugeniae* (**IPS:** *Cytospora eugeniae*)

In mature trees natural infection probably begins in **the absorbing roots spreading into larger roots and then to the collar.**

Valsa eugeniae can also cause a stem dieback functioning as **a wound parasite. Yellow stain is formed as in the root disease.**

Clove acute dieback - *Cryphonectria eugeniae*

Dark, reddish brown stain

Clove acute dieback - *Cryphonectria eugeniae*

Symptoms:

- The most noticeable symptom is the **death of a branch or a portion of a branch of the entire tree or in young saplings.**
- Symptoms usually include a progressive dieback of branches, leading to the main stem so as to produce **a stag-headed**‘ condition or even the death of the whole tree.
- If infection occurs near the base of the stem of young trees the whole plant will die suddenly with the leaves brown but still adhering to the branches.
- **Affected wood is sharply demarcated** from healthy tissues **by a dark, reddish brown stain**, there is profuse production of gum in the wood and the vessels are occluded by tyloses

Clove acute dieback - *Cryphonectria eugeniae*

Pathogen:

Teliomorph - *Endothia eugeniae* - Perithecia,

Perithecia are found on infected tissues at a later stage of the disease and airborne ascospores also contribute to spread of the disease

Anamorph - *Cryphonectria eugeniae* - Airborne conidia produced in pycnidia

Acute dieback occurs in trees older than 4-6 years.

It is **rare in younger trees and is never found in seedlings**

Clove acute dieback - *Cryphonectria eugeniae*

Management:

- Avoid causing damage to trees with machinery and tools.
- Infected parts of trees should be pruned out and burned.
- Any resultant wounds should be treated with Bordeaux paste.

Clove sumatra disease or die-back - *Ralstonia syzygii*

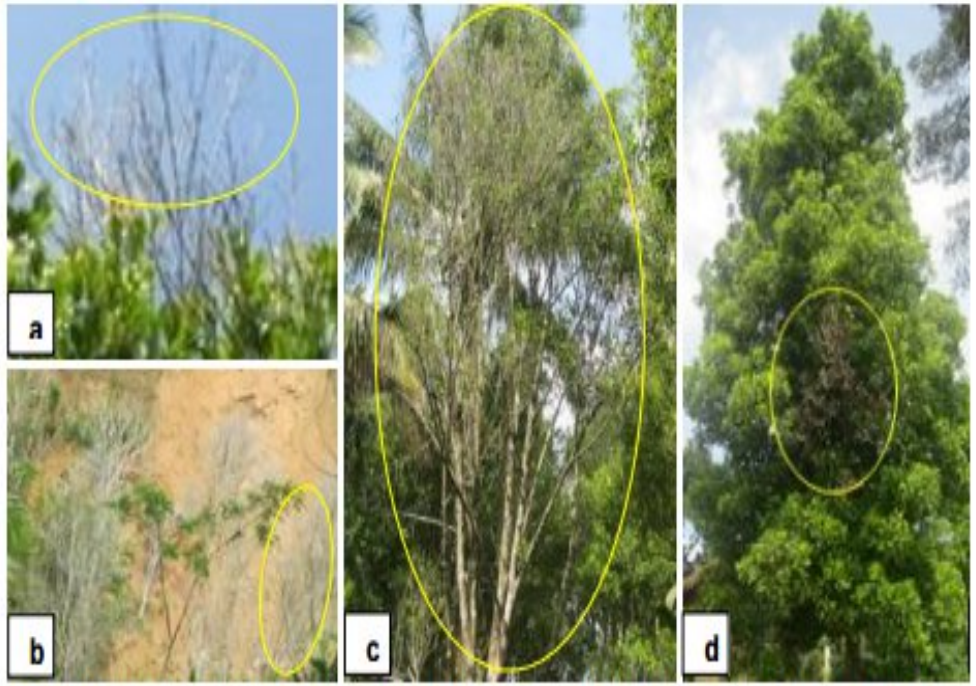

Fig. 1. Symptoms of wilt disease on the cloves. a. Autumn leaves start from the top. b. Plants become bald. c. Rare leafy plants. d. Dried leaves attached in a tree.

Clove sumatra disease or die-back - *Ralstonia syzygii*

Field symptoms. Pseudomonas syzygii, a disease of Cloves, Sumatra, July 1985. (J.M. Waller/CABI BioScience)

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Clove sumatra disease or die-back - *Ralstonia syzygii*

Clove sumatra disease or die-back - *Ralstonia syzygii*

- Sumatra disease, also known as Bakteria Pembuluh Kayu Cengkeh (BPKC: bacteria fill clove xylem) or Bacterial Vessels Wooden Cloves (BVWC)

Symptoms

- Diversity and spread of the disease may be caused by differences in **shade, temperature, sunlight, and season**.
- Early symptoms may be one-sided.
- The root system may show signs of general degeneration and decay.
- Dieback of trees which begins in the crown and leads to tree death within 3 years of initial infection

Clove sumatra disease or die back - *Ralstonia syzygii*

Symptoms contd…

- Leaves turn chlorotic and drop from the tree or may wilt suddenly and remain attached to the affected twigs/branches giving a scorched appearance
- Discoloration of vascular tissues evident as gray-brown streaks in new wood
- Bacterial exudate may ooze out of tissue when cut
- Symptoms of leaf-fall and twig die-back and often associated with wilting, lead to tree death in 6-18 months.
- Normally, only seedlings and young trees survive.
- Symptoms are usually more severe and disease spread is more rapid at higher elevation

Clove sumatra disease or die-back - *Ralstonia syzygii*

- Bacteria are limited to the water-carrying vessels in the tree (xylem)
- The disease is transmitted by *Hindola striata* and *Hindola fulva*, both sucking insects.

Clove sumatra disease or die back - *Ralstonia syzygii*

Management:

- An antibiotic - oxytetracycline - can be injected into the tree to slow the decline of infected trees
- Vector control (Hindola).
- Control of collateral weed host -Cleodendron

Diseases of cinnamon

1. Bark canker - *Phytophthora cinnamomi*
2. Leaf spot and die-back - *Colletotrichum gloeosporioides*

Cinnamon bark canker - *Phytophthora cinnamomi*

Figure 5: Rough bark disease on stems of cinnamon

Cinnamon bark canker - *Phytophthora cinnamomi*

- The bark canker of this crop occurs in the **west coast of sumatara and Indonesia**. The pathogen has also been reported from India.
- Stripe canker is found on the trunks and branches, particularly of young trees of *C. verum* or more generally on *C. burmannii* in Indonesia.
- ***P. cinnamomi* causing severe losses to forest trees and avocado**
- The fungus also affect ***C. campora, C. culitlauan* and *C. sintok***
- The fungus ***Phytophthora cinnamomi* from pineapple was also reported pathogenic on cinnamon, but with reduced virulence.**

Cinnamon bark canker - *Phytophthora cinnamomi*

Symptoms:

- Bark canker is found on the trunks and branches, particularly on young trees.
- **Vertical stripes are seen on the stems with amber colour exudates** at the advancing margins and hardens later.
- Vertical stripes of dead bark are most numerous near ground level.
- The disease is prevalent on **ill-drained soils and causes up to 42% damage.**

Cinnamon bark canker - *Phytophthora cinnamomi*

Pathogen:

- The hyphae are broad, hyphal swellings typically spherical (42 µm). Sporangiophore are thin (3 µm) and occasionally branched. Sporangia are broadly ellipsoidal or ovoid (33 x 57 µm), **papilla absent.**
- Oogonia (40 µm) have smooth walls and become yellow or golden with age.
- Antheredia amphigenous, long and measure 21-23 x 17 µm. Oospores hyaline, nearly filling the oogonium and thick.
- Fungus produces chlamydospores

Cinnamon bark canker - *Phytophthora cinnamomi*

Disease cycle and epidemiology

- The optimum temperature for growth of the fungus is 24-28° C.

Management:

- **Soil application of sulphur** has been recommended for the control of this disease.

Cinnamon leaf spot and die-back - *Colletotrichum gloeosporioides*

Leaf spot in Cinnamon leaves

Figure 4: Leaf spot symptoms in *Cinnamomum zeylanicum* caused by *Colletotrichum gloeosporioides*

Cinnamon leaf spot and die-back - *Colletotrichum gloeosporioides*

A severe **leaf spot and die-back disease has been observed in many parts of Kerala and serious both in nurseries and grownup trees.**

Symptoms:

- The symptoms in young seedlings include small brown specks on leaf lamina which later coalesce to form irregular patches.
- These small specks coalesce to form large necrotic blotches giving scorching appearance.
- Small, brownish, leaf spots on older leaves can be seen specially in shaded areas.
- Ultimately the leaves dry up and the pathogen spreads to the stem resulting in die-back symptoms.

Cinnamon leaf spot and die-back - *Colletotrichum gloeosporioides*

Symptoms contd…

- Spots later **become papery with dark brown margins**.
- **Light to deep brown concentric zones develop on the spots** during alternate dry and wet conditions.
- In other cases the central portion is shed, forming shot hole appearance.
- In severe infections, lesions may be larger than the half of the leaf.
- In India, reddish elongated spots were observed arising from the margin and resulting in defoliation.
- **Young seedlings are killed**. In mature trees, small specks appear which increase in size resulting in necrotic blotches.

Cinnamon leaf spot and die-back - *Colletotrichum gloeosporioides*

Pathogen:

- The disease is caused by fungus *Colletotrichum gloeosporioides*.
- Pathogen, disease cycle and epidemiology have been described under clove diseases

Management:

- Collect and destroy the infected plant debris.
- With the initiation of the disease, spray the crop with Bordeaux mixture (5:5:50) or copper oxychloride (0.3%) or combination of mancozeb (0.25%) and carbendazim (0.1%) and repeat at 14 days interval.

Diseases of nutmeg

1. Die back and fruit rot - *Phytophthora* spp. and *Diplodia natalensis*
2. Wilt - *Rosellinia pepo*
3. Leaf spot and shot hole - *Colletotrichum gloeosporioides*

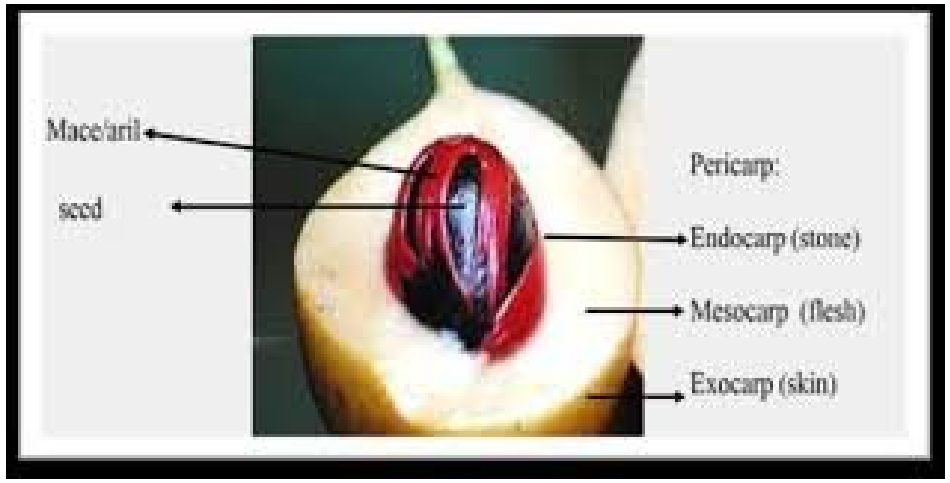
Nut meg die back and fruit rot -*Phytophthora* spp., *Diplodia natalensis*

Mature

Immature

The disease is characterized by drying up of mature and immature branches from the tip downwards.

Nut meg die back and fruit rot -*Phytophthora* spp., *Diplodia natalensis*



Rotting of mace and seed

Nut meg die back and fruit rot -*Phytophthora* spp., *Diplodia natalensis*

The disease can **be due to both physiological and pathological regions**.

Symptoms:

- The disease is characterized by drying up of **mature and immature branches** from the tip downwards.
- **Immature fruits split, rot and drop off** without apparent infection
- **In mature fruit, initially** water soaked dark lesions on pedicel later spreads to fruits and rind, which turns brown and rot due to disintegration of tissues.

Nut meg die back and fruit rot -*Phytophthora* spp., *Diplodia natalensis*

Symptoms contd..

- Brown discolouration of the rind is due to premature splitting of the pericarp and rotting of mace.
- In advanced stages, **the mace (seed cover) and seed also rots emitting a foul smell.**
- **The fallen fruits enveloped** by the fungal growth.

Diplodia spp

- Pycnidia black, single, globose, immersed, erumpent , papillate, ostiolate.
- Conidiophore/ conidiogenous cells- slender, hyaline, cylindrical
- Conidia-brown, 2 celled, ovoid , apex obtuse (round) and base truncate (shorten)

Diplodia

Nut meg die back and fruit rot -*Phytophthora* spp., *Diplodia natalensis*

Pathogen:

- The disease can be due to **both physiological and pathological regions.**
- The fungi associated with this disease are *Phytophthora* spp. and *Diplodia natalensis*

Management:

i) Spray the crop with Bordeaux mixture (5:5:50) when the fruits are half mature.

Nut meg wilt - *Rosellinia pepo*

Nut meg wilt - *Rosellinia pepo*

Symptoms

- Leaves of **badly infected trees wilt, turn brown, fall, and the tree eventually dies**
- **Fruits appear wrinkled and desiccated**.

Nut meg wilt - *Rosellinia pepo*

Management:

- The **infected branches should be** cut and removed and the cut ends pasted with 1% Bordeaux mixture or 0.2% copper oxychloride.

- Foliar spray of Pseudomonas at the rate of 20 g per litre of water.

Nut meg leaf spot and shot hole - *Colletotrichum gloeosporioides*

Sunken spots

Necrotic region

Nut meg leaf spot and shot hole - *Colletotrichum gloeosporioides*

Drying of leaves

Nut meg leaf spot and shot hole - *Colletotrichum gloeosporioides*

Symptoms:

- The initial symptoms of the disease appear as **sunken spots surrounded by a yellow halo.**
- Later, the central portion of the **necrotic region drops off resulting in shot hole symptoms.**
- **In some of the matured branches die back symptoms**.
- On young seedlings, **drying of the leaves and subsequent defoliation**

Nut meg leaf spot and shot hole - *Colletotrichum gloeosporioides*

Pathogen:

- Pathogen, disease cycle and epidemiology have already been described

Management:

i) Collect and destroy the infected plant debris.

ii) Apply prophylactic sprays of Bordeaux mixture (5:5:50) and repeat 2 or 3 times at 14 days interval

Root rot	Wilt
Predisposed by moisture stress	Predisposed by nematode infection also
Sudden death	Slow death
Bark shredding	No bark shredding
Disintegration of root system	No disintegration of root system
No browning of xylem vessels	Browning of xylem vessels
If pulled out, comes out very easily	Will not come easily
Flowering phase highly susceptible	All stage of crop growth
Ex-*Rhizoctonia bataticola, R.solani, S. rolfsii*	Ex- *Fusarium, Verticillum*